

The satellite polynomial is linear in the maximal-minor vector

MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors

theorem

Expresses the configuration polynomial as a linear pairing with maximal minors.

Statement

Let $n = m + 1 > 0$ and let J be finite, possibly empty. On $V = \mathbb{R}^n$ fix real linear functionals $F = (\ell_i)_{i=1}^n$ and $s = (s_a)_{a \in J}$. Put $M_F = (\ell_i(e_j))_{i,j}$ in the standard basis, and let $M_F[i \leftarrow s_a]$ replace row i by $(s_a(e_j))_j$. Set

$$A_{(F,s)}x = ((\ell_i(x))_i, (s_a(x))_a) \in \mathbb{R}^{n+|J|}.$$

Enumerate J to place its rows after the base rows. For a real linear map $A : V \rightarrow \mathbb{R}^{n+|J|}$, let $A_{ij} = (Ae_j)_i$. For each increasing n -row index I , put $\Pi_n(A)_I = \det(A_{I,*})$. Let I_0 select all base rows and $I_{i,a}$ select all base rows except i , followed by satellite a . Let e_I denote a minor-coordinate unit vector. Given any $w \in \mathbb{R}$ and coefficients $c_a \in V$, put

$$\Omega_{w,c} = we_{I_0} + \sum_{a \in J} \sum_{i=1}^n (c_a)_i (-1)^{n-i} e_{I_{i,a}},$$

$$\Phi_{w,c}(F, s) = w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a].$$

Then the finite dot product satisfies

$$\langle \Omega_{w,c}, \Pi_n(A_{(F,s)}) \rangle = \Phi_{w,c}(F, s).$$

Assumptions

The finite families, arbitrary real weight and coefficients are the only inputs. No norm model, admissibility or maximizing condition is assumed.

Conclusion

The same configuration appears in the map and polynomial. The parity corrects increasing-row order to row-replacement order.

Proof route

Read the row sign, expand both finite sums, and substitute their determinant identities.

▼ Proof steps

1. **Correct the selected row order.** Moving the last selected satellite row into position i crosses $n - i$ rows. Therefore

$$(-1)^{n-i} \Pi_n(A_{(F,s)})_{I_{i,a}} = \det M_F[i \leftarrow s_a], \quad \Pi_n(A_{(F,s)})_{I_0} = \det M_F.$$

The inverse cycle in the source maps position i to the last selected row. the parity-corrected minor lemma applies to this same configuration, i and a . In zero-based source indexing $i_0 = i - 1$, the parity is $(-1)^{m-i_0} = (-1)^{n-i}$. The pinned row-permutation determinant formula says $\det(M_{\sigma,*}) = \text{sign}(\sigma) \det M$; substitute the inverse cycle that moves the last selected row to i .

2. **Show both sums in the pairing.** Bilinearity gives

$$\begin{aligned} \langle \Omega_{w,c}, \Pi_n(A_{(F,s)}) \rangle &= w \Pi_n(A_{(F,s)})_{I_0} \\ &\quad + \sum_{a \in J} \sum_{i=1}^n (c_a)_i (-1)^{n-i} \Pi_n(A_{(F,s)})_{I_{i,a}} \\ &= w \det M_F + \sum_{a \in J} \sum_{i=1}^n (c_a)_i \det M_F[i \leftarrow s_a] \\ &= \Phi_{w,c}(F, s). \end{aligned}$$

The first equality expands the coefficient vector RHS; the second substitutes Step 1. If J is empty the sums vanish and the identity reduces to $w \det M_F$.

The product-indexed map and finite-coordinate map both send x to its base and satellite evaluations, as in the displayed definition of $A_{(F,s)}$. Extracting the rows of a linear map and reassembling them gives that same map, by the reconstruction identity.

For the later volume-scaled application, let p be a norm, $K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda(K_p)$ its reference Lebesgue volume. The volume-scaling identity gives

$$\langle \Omega_{w,c}, v_p \Pi_n(A_{(F,s)}) \rangle = v_p \Phi_{w,c}(F, s).$$

Neither the norm nor the volume factor is an input to the unscaled theorem above.

Main citations

- Exact declaration and source
- product-indexed map
- finite-coordinate map
- the reconstruction identity
- volume-scaling identity
- inverse cycle in the source
- the parity-corrected minor lemma
- row-permutation determinant formula
- the coefficient vector RHS

Lean source signature (exact)

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theorem pluckerPairing_satelliteCoefficients_maximalMinors
  (weight : ℝ) (coeff : J → Coord (m + 1))
  (C : SatelliteConfiguration (m + 1) J) :
  dotProduct (satellitePluckerCoefficients weight coeff)
    (maximalMinors (clmMatrix (configurationFinMap C))) =
    configurationPolynomial weight coeff C
```

In the source	Mathematical meaning
<code>m; [Fintype J]</code>	$n = m + 1 > 0$ and finite J , whose enumeration orders the satellite coordinates.
<code>C : SatelliteConfiguration (m + 1) J; C.1; C.2</code> <code>a</code>	The pair (F, s) ; $C.1$ is the base family (ℓ_i) and $C.2$ a is the functional s_a . The pair type imposes no admissibility.
<code>configurationFinMap C; clmMatrix</code> <code>(configurationFinMap</code> <code>C)</code>	The map $x \mapsto ((\ell_i(x))_i, (s_a(x))_a)$ and its matrix in standard coordinates.
<code>maximalMinors (clmMatrix (configurationFinMap</code> <code>C))</code>	The vector $\Pi_n(A_{(F,s)})$, with coordinate $\det(A_{I,*})$ for each increasing n -row index.
<code>replacementParity; baseMinorIndex;</code> <code>replacementMinorIndex</code>	The sign $(-1)^{n-i}$ and the row indices $I_0, I_{i,a}$; source index i_0 corresponds to formula index $i = i_0 + 1$.
<code>weight; coeff a; coeff a i</code>	w , the vector c_a , and its coordinate $(c_a)_i$, respectively.
In the source	Mathematical meaning
<code>satellitePluckerCoefficients weight coeff</code>	The full vector $\Omega_{w,c} = we_{I_0} + \sum_a \sum_i (c_a)_i (-1)^{n-i} e_{I_{i,a}}$.
<code>dotProduct (...) (...) = configurationPolynomial</code> <code>weight coeff</code> <code>C</code>	The whole equality $\langle \Omega_{w,c}, \Pi_n(A_{(F,s)}) \rangle = w \det M_F + \sum_a \sum_i (c_a)_i \det M_F[i \leftarrow s_a]$.

Surrounding assumptions and aliases.

▼ Exact content identity

Declaration: MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors
 Accepted content SHA-256: 6e0ca039827a0b6cb429f09633300fba57157e6ded5ee257f28f89a65861dcc9
 Accepted source guide SHA-256: f27d35366b349169534b09e4b62851a73ff429d3cef0c28c2b98fbb2eed96c5c
 Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73